

Telecom Customer Churn — Exploratory Data Analysis

Key Findings

1. Churn Distribution

Approximately **26.5% of customers churned**, while 73.5% were retained. The dataset is moderately imbalanced, which should be accounted for during modeling.

2. Tenure vs Churn

Customers who churned had significantly **lower tenure**. Most churned customers left within the first 12 months, while retained customers showed a wide spread across longer tenures suggesting early engagement is critical for retention.

3. Contract Type

Month-to-month contract holders form the **largest group** and have the highest churn rate. Customers on one-year or two-year contracts churn far less, indicating that locking customers into longer contracts is an effective retention strategy.

4. Internet & Add-on Services

Customers **without online security, online backup, device protection, or tech support** churned at noticeably higher rates compared to those subscribed to these services. This pattern held consistently across all four add on categories and suggests these services increase perceived value and stickiness.

Fiber optic internet users showed higher churn compared to DSL users, possibly due to pricing sensitivity or competitive alternatives.

5. Payment Method

Electronic check users had the highest churn among all payment methods. Customers on automatic payment methods (bank transfer or credit card) were significantly more likely to stay, possibly reflecting greater commitment and lower friction in billing.

Business Recommendations

Based on the patterns identified in this EDA, the following actions are recommended for the telecom business:

1. Prioritize Early-Tenure Customers

Since most churn happens within the first 12 months, the business should invest in an **onboarding program**, dedicated support, usage nudges, or welcome discounts, specifically targeting customers in their first quarter. Reducing early drop-off will have the highest impact on overall retention.

2. Incentivize Contract Upgrades

Month-to-month customers churn at a disproportionately high rate. Offering **upgrade incentives** (discounted first year, loyalty perks, or bundled services) to shift these customers to annual or two-year contracts should be a priority retention lever.

3. Bundle Add-on Services Proactively

Customers without online security, backup, or tech support are at significantly higher churn risk. Rather than treating these as upsells, consider bundling them into base plans or offering a **free trial period**, once customers experience the value, they are more likely to stay.

4. Investigate Fiber Optic Churn

Fiber optic users churning more than DSL users is a red flag, especially since fiber is a premium product. The business should investigate whether this is driven by **pricing, service quality, or competitor alternatives** and address the root cause before scaling fiber acquisition.

5. Migrate Electronic Check Users to Auto-Pay

Electronic check users show the highest churn. A targeted **auto-pay migration campaign**, with a small bill discount as an incentive — could reduce friction, improve payment reliability, and simultaneously lower churn risk for this segment.

6. Build a Churn Risk Score

The patterns identified here (low tenure + month-to-month + no add-ons + electronic check) form a natural **high-risk customer profile**. These variables should be used to build a simple churn risk score that the customer success or CRM team can act on proactively.